



Evaluation of seeding-competent alpha-synuclein in the CSF as a potential biomarker for early Parkinson's disease diagnosis and progression

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Summary

To prevent the early loss of affected brain cells in Parkinson's disease (PD) it will be critical to diagnose and treat patients as early as possible. Currently, no validated biomarkers for PD diagnosis or disease progression are available. Aggregation of the protein alpha-synuclein in the brain is strongly associated with the onset and progression of PD. The seeding aggregation assay (SAA) is a promising biomarker test for the detection of pathological, aggregation-inducing alpha-synuclein in the cerebrospinal fluid (CSF) from PD patients. The SAA biomarker test is therefore directly linked to the main pathological process in PD. We seek to answer the question if the SAA can be used for an accurate, early diagnosis of PD and if the SAA signal is linked to clinical features of the disease over time or in the future. We will measure SAA in CSF samples from 30 Parkinson's patients and 30 controls at the time of first clinical diagnosis and at follow-up three years later. In addition, we will analyze 20 CSF samples from atypical parkinsonian patients without dopaminergic deficiency (SWEDD). The accuracy of SAA for the diagnosis of PD versus controls and SWEDDs at the time of clinical diagnosis will be determined. Biomarker changes in individual patients over three years will be related to their disease progression.

Method

Human recombinant full-length (1-140 aa) alpha-synuclein without tag was expressed in *E. coli* and purified using standard purification methods. Alpha-synuclein from one single purification batch was used throughout this study. The SAA reaction buffer (RB) was composed of 100 mM phosphate buffer (pH 8.2), 10 μ M Thioflavin T (ThT) and 0.1 mg/ml alpha-synuclein. Each well of a black 96-well plate (Greiner bio-one, Germany) contained 95 μ l RB and 25 ± 3 mg of 0.1 mm zirconia/silica beads (Roth, Karlsruhe, Germany). Reactions were seeded with 5 μ l of undiluted CSF samples yielding a final reaction volume of 100 μ l per well. In addition, four AbbVie controls were run on each plate containing 5 μ l healthy control CSF and final concentrations of 0, 70, 700, and 7000 fg/ml preformed fibrils (PFFs) in a total reaction volume of 100 μ l per well. The plates were sealed with a plate sealer film (Fisher Scientific Ltd, UK) and incubated in a BMG FluoSTAR Omega plate reader at 33 °C for 70 h with shake/rest cycles: 1 min double orbital shaking at 200 rpm, 14 min rest. ThT fluorescence measurements (450 nm excitation and 480 nm emission; top





read) were taken every 15 min. Each CSF sample was measured in two independent assay runs with 4 technical replicates each. In total, 16 plates were measured each containing four AbbVie controls and two PPMI CSF pools in 4 technical replicates. In addition, plates 7, 8, 15, and 16 contained two ADB CSF pools in 4 technical replicates.

Before any further analysis, each aggregation curve was baseline corrected by subtracting the mean of the relative fluorescence units (rfu) measured between 2 and 5 h from all data points of the curve. A threshold of 3500 rfu was used to determine the time to threshold (TTT), i.e. the time when the curve reaches this threshold. If the threshold was not reached during the 70 h runtime of the assay, the TTT was set to 70 h. Curves that reached the threshold were defined as positive replicates. A sample was defined as SAA positive if 50% or more of the replicates were positive. In addition, the maximum fluorescence (Fmax) was determined as the highest rfu value recorded during the 70 h runtime of the assay. The area under curve (AUC) was calculated between 0 and 70 h.

Table 1: This table provides an overview of the terms used in the results file.

TESTNAME	Description
SAA Positive - final	Final result indicating if a CSF sample was SAA positive (1) or negative (0) based on the combined results from 2 independent runs (2 plates) with 4 technical replicates each. A sample was defined as SAA positive if 50% or more of the replicates were positive, i.e. 4 or more positive replicates out of 8 replicates in total. The rundate given in the results table corresponds to the second run (second plate).
SAA PR - plate [1-16]	Positive replicates (PR) per plate with 4 technical replicates in %.
SAA TTT - rep [1-8] plate [1-16]	Time to threshold (TTT) per plate and replicate in h. For each CSF sample there are 2 plates with 4 technical replicates each, i.e. a total of 8 replicates.
SAA Fmax - rep [1-8] plate [1-16]	Maximum fluorescence (Fmax) per plate and replicate in rfu. For each CSF sample there are 2 plates with 4 technical replicates each, i.e. a total of 8 replicates.
SAA AUC - rep [1-8] plate [1-16]	Area under curve (AUC) per plate and replicate. For each CSF sample there are 2 plates with 4 technical replicates each, i.e. a total of 8 replicates.
SMS_SUBJECT_ID (Control samples)	Description
PPMI-CSF-HC-POOL1	PPMI healthy control CSF pool. Run on all 16 plates in 4 technical replicates per plate.
PPMI-CSF-PD-POOL1	PPMI Parkinson's disease CSF pool. Run on all 16 plates in 4 technical replicates per plate.
ADB CSF alpha-syn high	ADB CSF pool with high alpha-synuclein level. Run on plates 7, 8, 15, and 16 in 4 technical replicates per plate.
ADB CSF alpha-syn low	ADB CSF pool with low alpha-synuclein level. Run on plates 7, 8, 15, and 16 in 4 technical replicates per plate.





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